

In-Frame Detection

BE 7910 Image Processing
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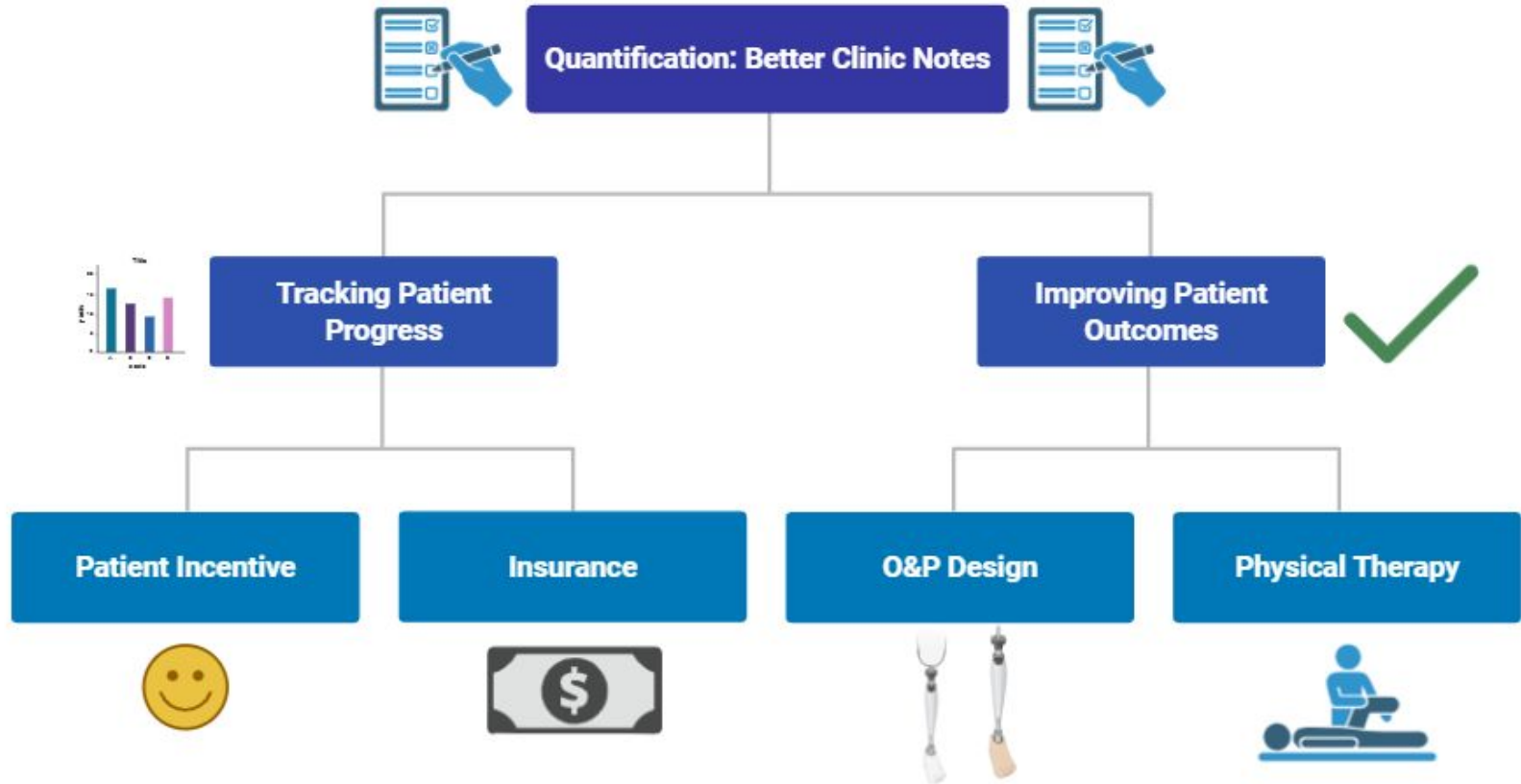
Introduction & Motivation

- ❖ My O&P experiences
 - Lack of detailed, quantitative data
 - Limitations of AMP scores



Figure 1. Ricki smoothing the inner rim of a transtibial socket at a Baton Rouge O&P clinic.

Impacts



Challenges

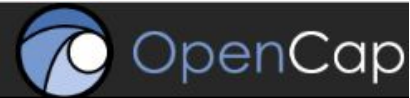
- ❖ Markered systems
- ❖  OpenCap not fully established as being trusted in general
 - Lacking for amputees
 - **One** study from 2024
- ❖ fit2stand.ai



Figure 2. OptiTrack motion capture cameras in LSU MIE's iCORE lab in PFT.

Master's Methodology

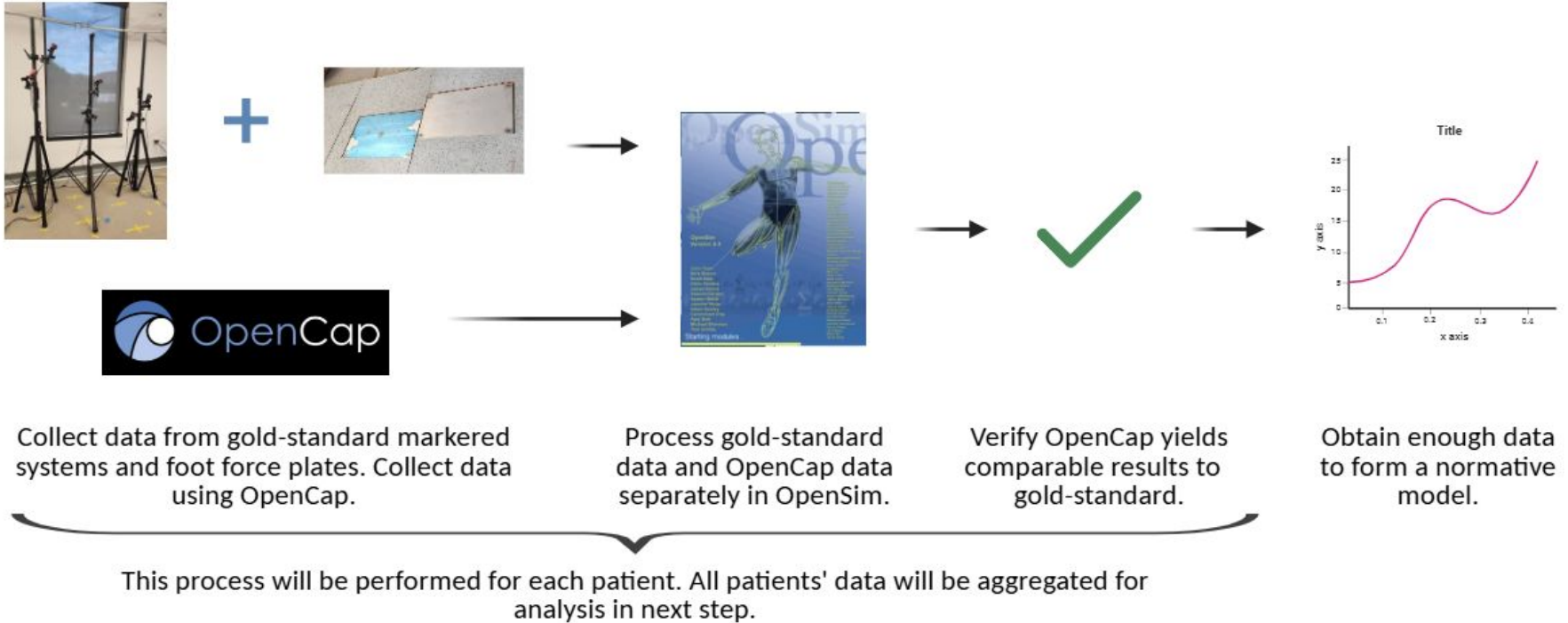


Figure 3. Proposed approach for my Master's project.

BE 7910's Methodology

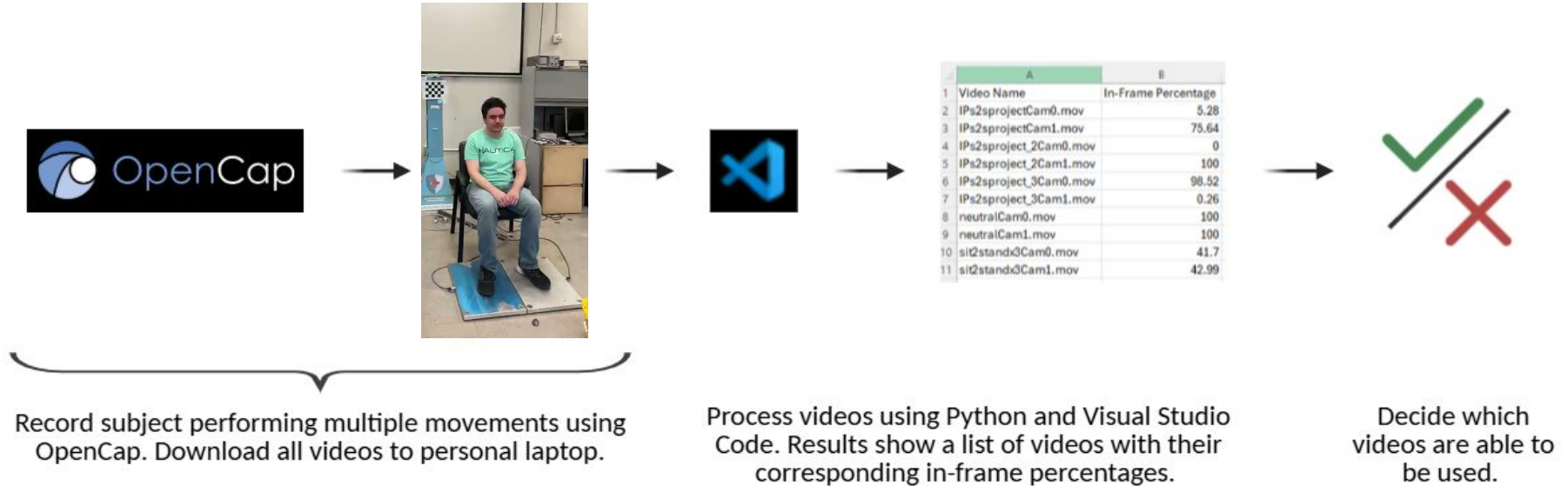


Figure 4. Proposed approach for BE 7910 project.

MediaPipe Pose

- ❖ Developed by Google in 2019
- ❖ Stage 1: Locates person
- ❖ Stage 2: uses CNN to estimate 33 landmarks
 - Drawn and connected

Python Pseudocode

- ❖ Import libraries and define path to videos
- ❖ Initialize MediaPipe Pose and define padding
- ❖ GetVideoMetadata
 - Use OpenCV to open video
 - Get resolution, FPS, frame count, duration
 - Use pymediainfo to get more detailed data

Python Pseudocode (continued)

- ❖ AnalyzeSubjectInFrame
 - Open video and initialize variables/counters
 - For all frames
 - Convert frames to RGB
 - Detect pose landmarks
 - Calculate and draw bounding box with padding
 - Record if bounding box is/is not in frame
 - Compute in-frame percentage
- ❖ Write results to .csv

Results

```
🎮 Processing video: IPs2sprojectCam0.mov

--- Metadata for IPs2sprojectCam0.mov ---
File size: 12.48 MB (12775.73 KB)
File format: MOV
Resolution: 1280 x 720
Frames per second (FPS): 59.98
Frame count: 720
Duration: 0:00:12
Codec: hvc1
Bit rate: 8704267 bps
Bit depth: 8 bit
Color space: YUV
Color range: Limited
Pixel aspect ratio: 1.000
Display aspect ratio: 1.778
Date: 2025-03-28 15:53:46 UTC

🧑 Subject fully in frame: 5.28% of frames.

🕒 Time intervals where subject was out of frame:
- From 0:00:00 to 0:00:02
- From 0:00:03 to 0:00:12
```

```
🎮 Processing video: IPs2sprojectCam1.mov

--- Metadata for IPs2sprojectCam1.mov ---
File size: 13.71 MB (14040.43 KB)
File format: MOV
Resolution: 1280 x 720
Frames per second (FPS): 59.98
Frame count: 780
Duration: 0:00:13
Codec: hvc1
Bit rate: 8830349 bps
Bit depth: 8 bit
Color space: YUV
Color range: Limited
Pixel aspect ratio: 1.000
Display aspect ratio: 1.778
Date: 2025-03-28 15:53:46 UTC

🧑 Subject fully in frame: 75.64% of frames.

🕒 Time intervals where subject was out of frame:
- From 0:00:00 to 0:00:00
- From 0:00:06 to 0:00:09
```

Figure 5. Snippet of results printed to Visual Studio Code interactive window. Results correspond to videos listed at top of image. These videos were downloaded as is from OpenCap.

Results



Figure 6. Panel showing original RGB image, landmark detection, landmarks connected, and bounding box from left to right. Notice this example does not include padding. Displayed frame #750 from the video `IPs2sprojectCam1.mov`.

Results

	A	B
1	Video Name	In-Frame Percentage
2	IPs2sprojectCam0.mov	5.28
3	IPs2sprojectCam1.mov	75.64
4	IPs2sproject_2Cam0.mov	0
5	IPs2sproject_2Cam1.mov	100
6	IPs2sproject_3Cam0.mov	98.52
7	IPs2sproject_3Cam1.mov	0.26
8	neutralCam0.mov	100
9	neutralCam1.mov	100
10	sit2standx3Cam0.mov	41.7
11	sit2standx3Cam1.mov	42.99

Figure 7. Example .csv file opened in Excel showing list of videos and percentage of time subject was in frame.

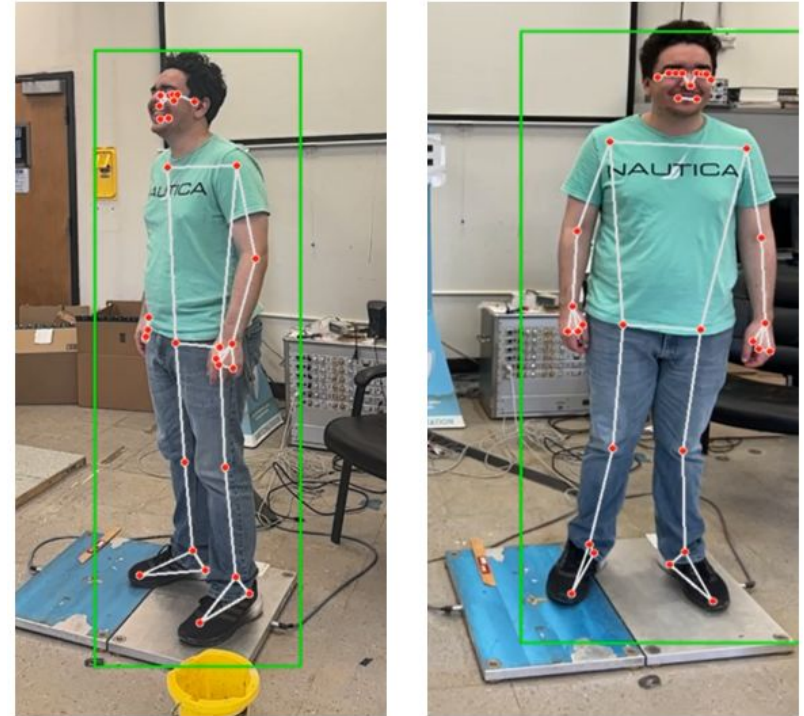
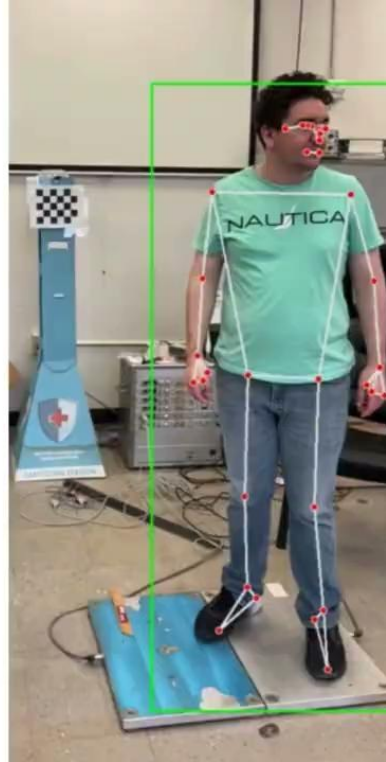


Figure 8. (Left) Landmark detection and bounding box when subject at angle vs (Right) subject facing camera.

Results



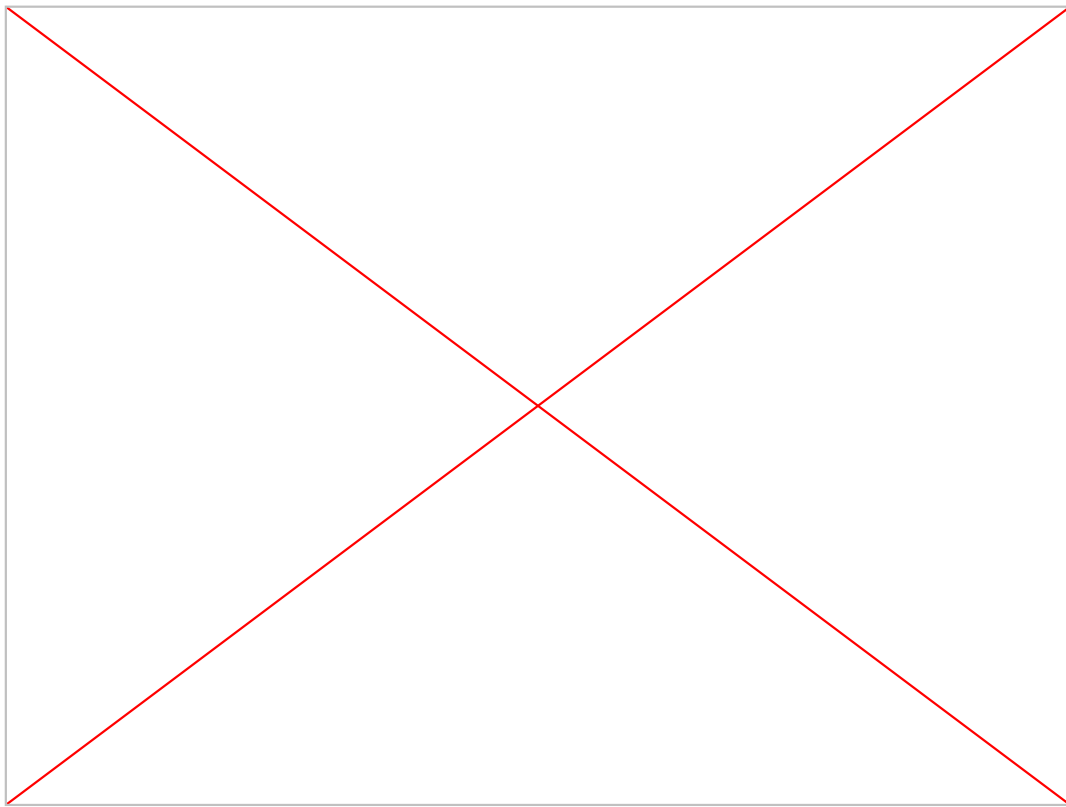
Video 1. Quick movements cause pose estimation to falter. Video showing how bounding box may not be best approach for edge detection. (Better to be safe than sorry...)

Discussion

❖ Limitations

- Bounding box based on skeletal landmarks rather than subject outline (MPP used to mimic markerless motion capture)
 - Padding hard coded
 - Slow
 - Use some frames
 - Parallel processing
 - Videos downloaded from OpenCap
- ## ❖ Traditional edge detection may increase accuracy

Lagniappe



Video 2. Detecting human outlines via Canny edge detector

Thank you!

Questions?

